



Aishu R 251001007-2025-IT ▾

A2**Started on** Monday, 24 August 2026, 12:28 PM**State** Finished**Completed on** Monday, 24 August 2026, 12:29 PM**Time taken** 1 min 55 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findFloor(int a[], int low, int high, int x)
4  {
5      int mid;
6
7      if (low > high)
8          return -1;
9
10     mid = (low + high) / 2;
11
12     if (a[mid] == x)
13         return a[mid];
14
15     if (a[mid] > x)
16         return findFloor(a, low, mid - 1, x);
17
18     {
19         int result = findFloor(a, mid + 1, high, x);
20
21         if (result == -1)
22             return a[mid];
23
24         return result;
25     }
26 }
27
28 int main()
29 {
30     int n, x, i, result;
31     int a[1000];
32
33     scanf("%d", &n);
34
35     for (i = 0; i < n; i++)
36         scanf("%d", &a[i]);
37
38     scanf("%d", &x);
39
40     result = findFloor(a, 0, n - 1, x);
41
42     printf("%d", result);
43
44     return 0;
45 }
46
```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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